

THE PROBLEM

Ghost workers — fictitious government employees collecting real salaries — cost Nigeria billions annually. Manual document verification is slow, inconsistent, and catches fraud too late.

23,846 ghost workers removed, Kaduna 2015

₦36.5B recovered in one audit year

30–40% of wage bill lost in affected agencies

THE USER

HR Verification Officer

Lagos State Ministry of Education

Manual verification takes 3-5 days per cycle. We check hundreds of documents by hand and still miss anomalies when payroll deadlines get tight.

Government Teacher × 2

Ogun State — Odeda & Ijebu-Ode LGAs

Some names on payroll do not correspond to anyone physically present at the school. Verification needs to happen before salaries move.

Field interviews conducted May 2026 — Lagos and Ogun State

THE SOLUTION

Lattice is an AI-powered ghost worker detection SDK that verifies every government employee is real before their salary is disbursed through Squad.

- **Liveness Detection**
MediaPipe Face Mesh — confirms a live human is present via challenge-response
- **Deepfake Classification**
EfficientNet-B0 — catches AI-generated and manipulated faces before BVN check
- **Anomaly Detection**
Isolation Forest + SHAP — flags ghost worker clusters across the full payroll cohort

Output — Verifiable Institutional Quote (VIQ)

```
{
  "trust_score": 94,
  "verdict": "PASS",
  "status": "PAID_AND_VERIFIED"
}
```

Signed. Auditable. Permanent. One record per worker per pay cycle.

THE FLOW

1. Worker receives Squad SMS OTP
2. Completes liveness challenge on device
3. Deepfake classifier screens captured frame
4. BVN verified via Squad endpoint
5. Isolation Forest scans full payroll cohort
6. Trust Score generated — VIQ signed
7. Score ≥ 80: Squad transfer fires automatically

< 8 seconds: liveness to VIQ — < 30 seconds: VIQ to payment confirmed

SQUAD APIS USED

SMS API

Delivers OTP challenge to worker at session start

BVN Verification

Resolves legal name and date of birth — cross-checked against payroll record

Virtual Accounts

One persistent account per worker — full salary audit trail

Conditional Transfer

Payment fires only when Trust Score ≥ 80 — Squad enforces the gate

Webhooks

PAID_AND_VERIFIED confirmation closes the audit loop permanently

THE FOUR PILLARS

- **AI Automation**
Three distinct AI models working in sequence — no human needed for standard verification
- **Use of Data**
Behavioral anomaly detection on payroll cohort — SHAP explainability for every flag
- **Squad APIs**
Five endpoints — SMS, BVN, virtual accounts, transfers, webhooks — all mission-critical
- **Financial Innovation**
Trust-gated disbursement — salary only moves when identity is verified

IMPACT

300,000+ Lagos civil servants, Year 1 reach

₦15M/mo revenue at Lagos State scale (₦50/VIQ)

160x ROI fraud prevention vs verification cost

BIAS AUDIT

Evaluated on Fitzpatrick Scale IV–VI (Nigerian skin tones). False Rejection Rate differential < 2 percentage points across all groups. Full methodology in GitHub README.